

Hospital Management System

Database Design Document

Rough Schema · Primary & Foreign Keys · Sample Dummy Data

20 Core Tables covering Patients, Doctors, Appointments,
Queue Management, Prescriptions, Pharmacy, Payments & Audit

- PK = Primary Key
- FK = Foreign Key (references another table)

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01 USERS

System login accounts (admin, doctor, reception, pharmacy, lab staff).

Column	Data Type	Key	Reference
user_id	INT	PK	-
username	VARCHAR(50)		-
password	VARCHAR(255)		-
role	ENUM		-
status	ENUM(Active/Inactive)		-
created_at	DATETIME		-

Sample dummy data:

user_id	username	password	role	status	created_at
1	dr.mehta	*****hash1	Doctor	Active	2025-01-10 09:00
2	reception1	*****hash2	Receptionist	Active	2025-01-11 10:15
3	admin	*****hash3	Admin	Active	2025-01-01 08:00

02 DEPARTMENTS

Hospital departments / specialties, referenced by DOCTORS.

Column	Data Type	Key	Reference
department_id	INT	PK	-
department_name	VARCHAR(100)		-
floor	VARCHAR(20)		-
description	TEXT		-

Sample dummy data:

department_id	department_name	floor	description
1	Cardiology	2nd Floor	Heart & vascular care
2	Pediatrics	1st Floor	Child healthcare
3	Orthopedics	3rd Floor	Bone & joint care

03 DOCTORS

Doctor master records; belongs to a department.

Column	Data Type	Key	Reference
doctor_id	INT	PK	-
name	VARCHAR(100)		-
specialization	VARCHAR(100)		-
department_id	INT	FK	DEPARTMENTS.department_id

Column	Data Type	Key	Reference
qualification	VARCHAR(100)		-
experience	INT (years)		-
phone	VARCHAR(15)		-
email	VARCHAR(100)		-
room_no	VARCHAR(10)		-
availability	VARCHAR(100)		-
consultation_fee	DECIMAL(8,2)		-

Sample dummy data:

doctor_id	name	specialization	department_id	qualification	experience	phone	email	room_no	availability	consultation_fee
1	Dr. R. Mehta	Cardiologist	1	MD, DM Cardio	12	9876500001	mehta@hospital.in	201	Mon-Sat 10-2	500.00
2	Dr. S. Iyer	Pediatrician	2	MD Peds	8	9876500002	iyer@hospital.in	105	Mon-Fri 9-1	350.00
3	Dr. A. Shah	Orthopedic	3	MS Ortho	15	9876500003	shah@hospital.in	301	Tue-Sun 11-3	450.00

04 PATIENTS

Core patient master / demographic record.

Column	Data Type	Key	Reference
patient_id	INT	PK	-
health_id	VARCHAR(20)		-
first_name	VARCHAR(50)		-
last_name	VARCHAR(50)		-
gender	ENUM(M/F/O)		-
dob	DATE		-
blood_group	VARCHAR(5)		-
phone	VARCHAR(15)		-
email	VARCHAR(100)		-
address	VARCHAR(200)		-
city	VARCHAR(50)		-
state	VARCHAR(50)		-
pincode	VARCHAR(10)		-
aadhaar	VARCHAR(20)		-
abha_id	VARCHAR(20)		-
height	DECIMAL(5,2) cm		-
weight	DECIMAL(5,2) kg		-

Column	Data Type	Key	Reference
emergency_contact	VARCHAR(15)		-
registration_date	DATE		-

Sample dummy data:

pati ent_ id	heal th_i d	first _na me	last_ nam e	gen der	dob	bloo d_gr oup	pho ne	emai l	addr ess	city	stat e	pinc ode	aad haar	abh a_id	heig ht	weig ht	eme rgen cy_c onta ct	regi strat ion_ date
101	HID-1001	Aarav	Patel	M	1990-04-12	B+	9998811001	aarav.p@gmail.com	12 MG Road	Ahmedabad	Gujarat	380001	xxxx-xxx-x-1001	ABHA-1001	172.0	70.5	9998811999	2024-06-01
102	HID-1002	Isha	Shah	F	1985-11-02	O+	9998811002	isha.s@mail.com	45 Ring Road	Ahmedabad	Gujarat	380015	xxxx-xxx-x-1002	ABHA-1002	160.0	58.0	9998811998	2024-07-15
103	HID-1003	Kabir	Rao	M	2018-02-20	A+	9998811003	-	9 Nehru Nagar	Ahmedabad	Gujarat	380009	xxxx-xxx-x-1003	ABHA-1003	110.0	20.0	9998811997	2025-01-05

05 APPOINTMENTS

Booking record linking a patient with a doctor for a visit.

Column	Data Type	Key	Reference
appointment_id	INT	PK	-
patient_id	INT	FK	PATIENTS.patient_id
doctor_id	INT	FK	DOCTORS.doctor_id
appointment_date	DATE		-
appointment_time	TIME		-
status	ENUM(Booked/Completed/Cancelled)		-
token_number	INT		-
priority	VARCHAR(20)		-
booking_type	ENUM(Online/Walk-in)		-
remarks	VARCHAR(200)		-

Sample dummy data:

appointme nt_id	patient_id	doctor_id	appointme nt_date	appointme nt_time	status	token_nu mber	priority	booking_t ype	remarks
5001	101	1	2025-07-08	10:30	Booked	12	Normal	Online	Follow-up checkup

appointment_id	patient_id	doctor_id	appointment_date	appointment_time	status	token_number	priority	booking_type	remarks
5002	102	2	2025-07-08	11:00	Completed	5	Senior Citizen	Walk-in	-
5003	103	3	2025-07-09	09:15	Booked	2	Child	Online	Knee pain review

06 TOKENS

Token issued for each appointment for queue tracking.

Column	Data Type	Key	Reference
token_id	INT	PK	-
appointment_id	INT	FK	APPOINTMENTS.appointment_id
token_number	INT		-
queue_position	INT		-
estimated_wait_time	INT (mins)		-
status	ENUM(Waiting/Serving/Done)		-

Sample dummy data:

token_id	appointment_id	token_number	queue_position	estimated_wait_time	status
9001	5001	12	3	25	Waiting
9002	5002	5	0	0	Done
9003	5003	2	1	10	Waiting

07 QUEUE

Live doctor-wise patient queue, ordered by priority level.

Column	Data Type	Key	Reference
queue_id	INT	PK	-
patient_id	INT	FK	PATIENTS.patient_id
doctor_id	INT	FK	DOCTORS.doctor_id
priority_level	ENUM (see priority list)		-
queue_number	INT		-
arrival_time	DATETIME		-
served_time	DATETIME		-
status	ENUM(Waiting/In-progress/Served)		-

Sample dummy data:

queue_id	patient_id	doctor_id	priority_level	queue_number	arrival_time	served_time	status
701	103	3	Child	1	2025-07-09 09:00	-	Waiting
702	102	2	Senior Citizen	2	2025-07-08 10:50	2025-07-08 11:05	Served
703	101	1	Normal	3	2025-07-08 10:20	-	Waiting

08 PRESCRIPTIONS

Diagnosis and notes recorded for a patient visit.

Column	Data Type	Key	Reference
prescription_id	INT	PK	-
patient_id	INT	FK	PATIENTS.patient_id
doctor_id	INT	FK	DOCTORS.doctor_id
visit_id	INT	FK	APPOINTMENTS.appointment_id
diagnosis	VARCHAR(200)		-
notes	TEXT		-
created_date	DATE		-

Sample dummy data:

prescription_id	patient_id	doctor_id	visit_id	diagnosis	notes	created_date
301	101	1	5001	Hypertension - stable	Continue current medication	2025-07-08
302	102	2	5002	Viral fever	Rest & hydration advised	2025-07-08
303	103	3	5003	Mild knee sprain	Physiotherapy recommended	2025-07-09

09 PRESCRIPTION_MEDICINES

Medicines & dosage schedule attached to a prescription.

Column	Data Type	Key	Reference
id	INT	PK	-
prescription_id	INT	FK	PRESCRIPTIONS.prescription_id
medicine_name	VARCHAR(100)		-
dosage	VARCHAR(50)		-
morning	BOOLEAN		-
afternoon	BOOLEAN		-
night	BOOLEAN		-

Column	Data Type	Key	Reference
days	INT		-

Sample dummy data:

id	prescription_id	medicine_name	dosage	morning	afternoon	night	days
1	301	Amlodipine 5mg	1 tablet	Yes	No	No	30
2	302	Paracetamol 500mg	1 tablet	Yes	Yes	Yes	5
3	303	Ibuprofen 400mg	1 tablet	No	Yes	Yes	7

10 MEDICAL_HISTORY

Past diseases, allergies, surgeries and family history.

Column	Data Type	Key	Reference
history_id	INT	PK	-
patient_id	INT	FK	PATIENTS.patient_id
disease	VARCHAR(100)		-
allergy	VARCHAR(100)		-
surgery	VARCHAR(100)		-
family_history	VARCHAR(200)		-
description	TEXT		-

Sample dummy data:

history_id	patient_id	disease	allergy	surgery	family_history	description
401	101	Hypertension	None	Appendectomy (2015)	Father - Diabetic	Diagnosed 2019
402	102	None	Penicillin	None	Mother - Hypertension	-
403	103	Asthma (mild)	Dust	None	None	Since early childhood

11 LAB_TESTS

Lab test orders and report status per patient.

Column	Data Type	Key	Reference
test_id	INT	PK	-
patient_id	INT	FK	PATIENTS.patient_id
doctor_id	INT	FK	DOCTORS.doctor_id
test_name	VARCHAR(100)		-

Column	Data Type	Key	Reference
status	ENUM(Pending/Completed)		-
test_date	DATE		-
report_file	VARCHAR(255) (path/URL)		-

Sample dummy data:

test_id	patient_id	doctor_id	test_name	status	test_date	report_file
501	101	1	Lipid Profile	Completed	2025-07-05	/reports/501.pdf
502	102	2	CBC	Pending	2025-07-08	-
503	103	3	X-Ray Knee	Completed	2025-07-09	/reports/503.pdf

12 VACCINATIONS

Vaccination records and next-due tracking.

Column	Data Type	Key	Reference
vaccination_id	INT	PK	-
patient_id	INT	FK	PATIENTS.patient_id
vaccine_name	VARCHAR(100)		-
date	DATE		-
next_due	DATE		-

Sample dummy data:

vaccination_id	patient_id	vaccine_name	date	next_due
601	103	DTP Booster	2025-02-01	2026-02-01
602	101	Influenza	2025-01-15	2026-01-15
603	102	Tetanus Toxoid	2024-12-01	2029-12-01

13 FOLLOW_UP

Scheduled follow-up visits recommended by a doctor.

Column	Data Type	Key	Reference
followup_id	INT	PK	-
patient_id	INT	FK	PATIENTS.patient_id
doctor_id	INT	FK	DOCTORS.doctor_id
next_visit	DATE		-
remarks	VARCHAR(200)		-
status	ENUM(Pending/Done)		-

Sample dummy data:

followup_id	patient_id	doctor_id	next_visit	remarks	status
701	101	1	2025-08-08	Review BP readings	Pending
702	103	3	2025-07-23	Physiotherapy progress check	Pending
703	102	2	2025-07-15	Confirm fever resolved	Done

14 REMINDERS

Generic reminders sent to patients (appointment, medicine, vaccine).

Column	Data Type	Key	Reference
reminder_id	INT	PK	-
patient_id	INT	FK	PATIENTS.patient_id
type	ENUM(Appointment/Medicine/Vaccine)		-
date	DATETIME		-
status	ENUM(Sent/Pending)		-

Sample dummy data:

reminder_id	patient_id	type	date	status
801	101	Appointment	2025-08-07 09:00	Pending
802	103	Vaccine	2026-01-25 09:00	Pending
803	102	Medicine	2025-07-13 08:00	Sent

15 DIET_PLAN

Doctor-recommended diet plan for a patient.

Column	Data Type	Key	Reference
diet_id	INT	PK	-
patient_id	INT	FK	PATIENTS.patient_id
doctor_id	INT	FK	DOCTORS.doctor_id
breakfast	VARCHAR(200)		-
lunch	VARCHAR(200)		-
snacks	VARCHAR(200)		-
dinner	VARCHAR(200)		-
water	VARCHAR(50)		-
instructions	TEXT		-

Sample dummy data:

diet_id	patient_id	doctor_id	breakfast	lunch	snacks	dinner	water	instructions
901	101	1	Oats + fruit	Low-salt thali	Nuts	Grilled veg + roti	2.5 L	Avoid fried food
902	102	2	Fruit juice + toast	Khichdi	Buttermilk	Soup + toast	2 L	Light meals until fever subsides

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PHARMACY

Medicine stock master used by the in-house pharmacy.

Column	Data Type	Key	Reference
medicine_id	INT	PK	-
medicine_name	VARCHAR(100)		-
stock	INT		-
expiry	DATE		-
price	DECIMAL(8,2)		-
manufacturer	VARCHAR(100)		-

Sample dummy data:

medicine_id	medicine_name	stock	expiry	price	manufacturer
1	Amlodipine 5mg	500	2027-03-01	2.50	Cipla
2	Paracetamol 500mg	1200	2026-11-15	1.00	Sun Pharma
3	Ibuprofen 400mg	800	2026-09-30	1.75	Zydus

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MEDICINE_ISSUE

Medicines dispensed to a patient from PHARMACY stock.

Column	Data Type	Key	Reference
issue_id	INT	PK	-
patient_id	INT	FK	PATIENTS.patient_id
medicine_id	INT	FK	PHARMACY.medicine_id
quantity	INT		-
issue_date	DATE		-

Sample dummy data:

issue_id	patient_id	medicine_id	quantity	issue_date
1001	101	1	30	2025-07-08
1002	102	2	10	2025-07-08
1003	103	3	14	2025-07-09

18 PAYMENTS

Payment transactions for consultations, tests, or medicines.

Column	Data Type	Key	Reference
payment_id	INT	PK	-
patient_id	INT	FK	PATIENTS.patient_id
appointment_id	INT	FK	APPOINTMENTS.appointment_id
amount	DECIMAL(10,2)		-
payment_mode	ENUM(Cash/Card/UPI)		-
status	ENUM(Paid/Pending/Refunded)		-
date	DATE		-

Sample dummy data:

payment_id	patient_id	appointment_id	amount	payment_mode	status	date
2001	101	5001	500.00	UPI	Paid	2025-07-08
2002	102	5002	350.00	Cash	Paid	2025-07-08
2003	103	5003	450.00	Card	Paid	2025-07-09

19 FEEDBACK

Patient feedback and rating for a doctor / visit.

Column	Data Type	Key	Reference
feedback_id	INT	PK	-
patient_id	INT	FK	PATIENTS.patient_id
doctor_id	INT	FK	DOCTORS.doctor_id
rating	INT (1-5)		-
comments	TEXT		-
date	DATE		-

Sample dummy data:

feedback_id	patient_id	doctor_id	rating	comments	date
3001	101	1	5	Very attentive doctor	2025-07-08
3002	102	2	4	Short wait, good advice	2025-07-08
3003	103	3	5	Explained the treatment clearly	2025-07-09

20 LOGIN_HISTORY

Login/logout audit trail per user account.

Column	Data Type	Key	Reference
login_id	INT	PK	-
user_id	INT	FK	USERS.user_id
login_time	DATETIME		-
logout_time	DATETIME		-
ip_address	VARCHAR(45)		-

Sample dummy data:

login_id	user_id	login_time	logout_time	ip_address
1	1	2025-07-08 09:00	2025-07-08 14:05	192.168.1.11
2	2	2025-07-08 08:45	2025-07-08 17:30	192.168.1.22
3	3	2025-07-08 08:00	2025-07-08 18:10	192.168.1.5

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AUDIT_LOG

System-wide audit trail of create/update/delete operations.

Column	Data Type	Key	Reference
log_id	INT	PK	-
user	VARCHAR(50)	FK	USERS.username
operation	ENUM(INSERT/UPDATE/DELETE)		-
table_name	VARCHAR(50)		-
timestamp	DATETIME		-

Sample dummy data:

log_id	user	operation	table_name	timestamp
1	reception1	INSERT	APPOINTMENTS	2025-07-08 10:29
2	dr.mehta	UPDATE	PRESCRIPTIONS	2025-07-08 10:45
3	admin	UPDATE	PHARMACY	2025-07-08 11:00

Priority Queue — Ordering Levels

Patients in the QUEUE table are served according to the following priority order (highest priority first):

Rank	Priority Level
1	Emergency
2	Pregnant
3	Senior Citizen
4	Child
5	Disabled
6	Normal

Entity Relationship Summary

High-level relationships between the core tables (rough draft — PATIENTS and DOCTORS are the two central hubs the rest of the schema branches from):

From	Type	To	Description
PATIENTS	1 : N	APPOINTMENTS	One patient can have many appointments
DOCTORS	1 : N	APPOINTMENTS	One doctor attends many appointments
DEPARTMENTS	1 : N	DOCTORS	One department has many doctors
APPOINTMENTS	1 : 1	TOKENS	Each appointment gets one token
PATIENTS	1 : N	QUEUE	A patient can appear in the live queue
DOCTORS	1 : N	QUEUE	A doctor's queue lists many patients
PATIENTS	1 : N	PRESCRIPTIONS	One patient has many prescriptions
PRESCRIPTIONS	1 : N	PRESCRIPTION_MEDICINES	One prescription has many medicine lines
PATIENTS	1 : N	MEDICAL_HISTORY	One patient has many history entries
PATIENTS	1 : N	LAB_TESTS	One patient can have many lab tests
PATIENTS	1 : N	VACCINATIONS	One patient has many vaccination records
PATIENTS	1 : N	FOLLOW_UP	One patient has many follow-ups
PATIENTS	1 : N	REMINDERS	One patient has many reminders
PATIENTS	1 : N	DIET_PLAN	One patient can have diet plans
PHARMACY	1 : N	MEDICINE_ISSUE	One medicine is issued many times
PATIENTS	1 : N	PAYMENTS	One patient makes many payments
PATIENTS	1 : N	FEEDBACK	One patient can give feedback per visit
USERS	1 : N	LOGIN_HISTORY	One user has many login sessions

Note: this is a rough / draft schema intended to communicate structure and relationships. Field lengths, exact data types, indexes, and additional constraints (NOT NULL, UNIQUE, CHECK) should be finalized with the development team before implementation.